

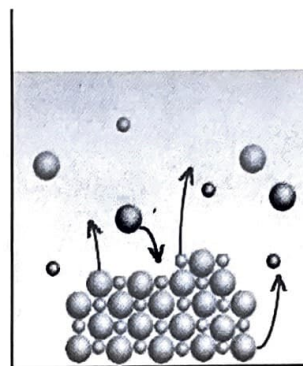
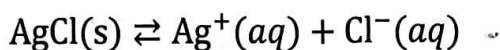
Solubility Equilibria: Dissolve All Your Troubles Away!

Up until now, we've pretended that compounds fall into one of two categories: 100% soluble or 100% insoluble. Actually, an equilibrium can exist between a partially soluble substance and its solution.

Solubility Product Constant (K_{sp})

Equilibrium expression for dissolving a solid into ions

Like all K values, K_{sp} is constant at a constant temperature.



When the rate at which a solid dissolves into ions is _____ to the rate at which ions precipitate back to solid, the system has reached equilibrium.

Let's Practice! Write the K_{sp} expression for each of the following dissolutions:

Salt	Dissociation reaction	K_{sp} Expression
K_2CO_3	$\text{K}_2\text{CO}_3(s) \rightleftharpoons 2\text{K}^+(aq) + \text{CO}_3^{2-}(aq)$	$K_{sp} = [\text{K}^+]^2 [\text{CO}_3^{2-}]$
Al_2S_3	$\text{Al}_2\text{S}_3(s) \rightleftharpoons 2\text{Al}^{3+}(aq) + 3\text{S}^{2-}(aq)$	$K_{sp} = [\text{Al}^{3+}]^2 [\text{S}^{2-}]^3$

Wait a sec, K_{sp} only has products ! But why?

Small K_{sp} ($K \ll 1$)	Larger K_{sp}	100% Soluble
- Only a small amount of solid dissociates into ions - Lower solubility	- More solid dissociates into ions - Higher solubility	- All solid dissociates into ions - Dissolves to completion

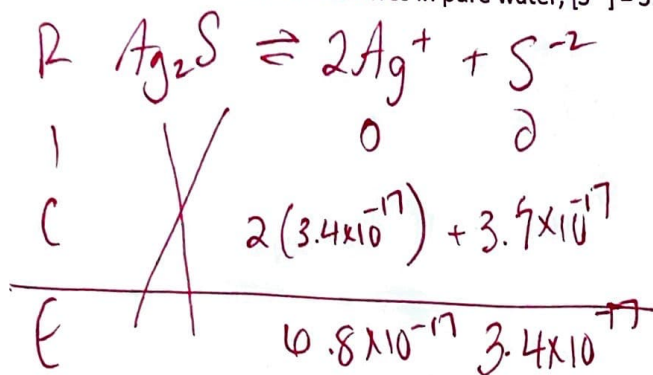
There are three types of 100% soluble ions that you have to memorize:

Always soluble: alkali metal cations, NH_4^+ , NO_3^-

K_{sp} Calculations

ALWAYS write the balanced dissociation reaction and K_{sp} expression before solving!

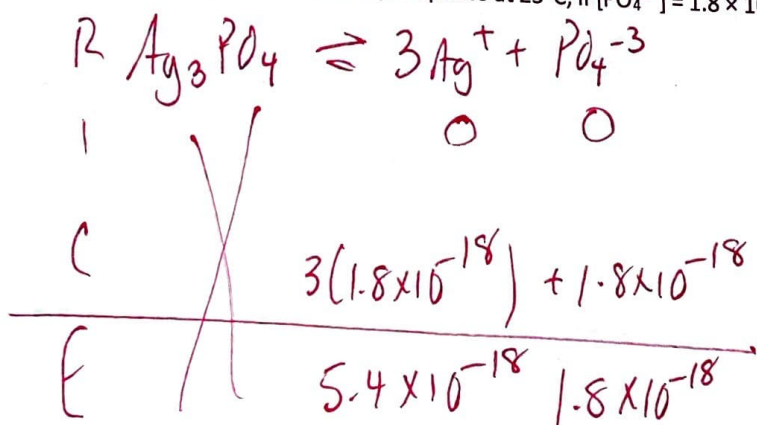
Example: When silver sulfide dissolves in pure water, $[S^{2-}] = 3.4 \times 10^{-17}$ mol/L at 25°C. Calculate its K_{sp} value.



$$\begin{aligned}
 K_{sp} &= [Ag]^2 [S^{2-}] \\
 &= (6.8 \times 10^{-17})^2 \cdot (3.4 \times 10^{-17}) \\
 &= 1.6 \times 10^{-49}
 \end{aligned}$$

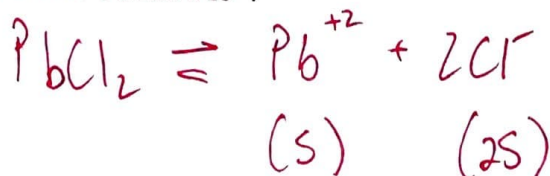
Now you try!

1. What is the K_{sp} value of silver phosphate at 25°C, if $[PO_4^{3-}] = 1.8 \times 10^{-18}$ M when dissolved in pure water?



$$\begin{aligned}
 K_{sp} &= [Ag^+]^3 [PO_4^{3-}] \\
 &= (5.4 \times 10^{-18})^3 \cdot (1.8 \times 10^{-18}) \\
 &= 9.4 \times 10^{-71}
 \end{aligned}$$

2. What is the concentration of chloride ions that will form in a solution of PbCl₂ in pure water? The K_{sp} of lead (II) chloride at 25°C is 1.17×10^{-5} .



$$\begin{aligned}
 K_{sp} &= [Pb^{+2}] [Cl^-]^2 \\
 &= (s) (2s)^2 \\
 K_{sp} &= 4s^3 = 1.17 \times 10^{-5} \\
 s &= 1.43 \times 10^{-2}
 \end{aligned}$$

$$[Cl^-] = 2s = 2.86 \times 10^{-2} M$$

But wait! The "S" term, for dissociation, always refers to the amount of Solute that will dissolve (since the stoichiometric coefficient of the salt will always be 1 in the dissociation reaction. This term has a special name!

Solubility "S" (aka Molar Solubility) = "x" in your K_{sp} RICE Table

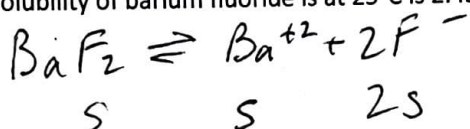
How much of a S will dissolve per one L of solution (Units: M = mol/L)

Solubility is an equilibrium position and therefore can change (for example, if you change the number of ions in solution, this will shift the equilibrium position and thus, the solubility).

- Larger molar solubility values suggest more dissociation into ions and greater solubility.
- Smaller molar solubility values suggest less dissociation into ions and lower solubility.
- **Note:** when comparing solubility of given compounds, you should compare their molar solubility values and NOT their K_{sp} values!

Now we can answer problems using the same math we used on the previous page. The terminology will be different, but the calculations are the same! Let's try some:

3. The molar solubility of barium fluoride is at 25°C is 2.45×10^{-5} M. Calculate K_{sp} .



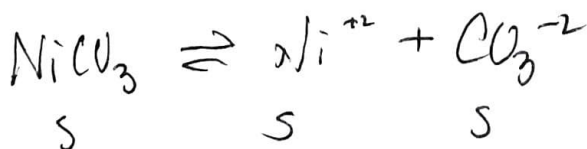
$$K_{sp} = [\text{Ba}^{+2}][\text{F}^{-}]^2$$

$$= \text{S} \cdot (2\text{S})^2 = 4\text{S}^3$$

$$K_{sp} = 4 \cdot (2.45 \times 10^{-5})^3$$

$$=$$

4. Calculate the molar solubility of nickel (II) carbonate, which has a K_{sp} of 1.4×10^{-7} at 25°C.



$$K_{sp} = [\text{Ni}^{+2}][\text{CO}_3^{-2}]$$

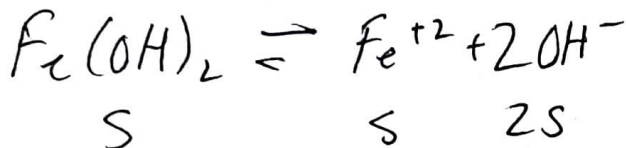
$$= \text{S}^2 = 1.4 \times 10^{-7}$$

$$\text{S} = 3.7 \times 10^{-4}$$

How Much Will Dissolve?

You can use molar solubility to determine how many grams of a solid will dissolve in a quantity of water!

5. How many grams of iron (II) hydroxide can dissolve in 500. mL of water? Its $K_{sp} = 4.87 \times 10^{-17}$ at 25°C.



$$K_{sp} = [\text{Fe}^{+2}][\text{OH}^-]^2$$

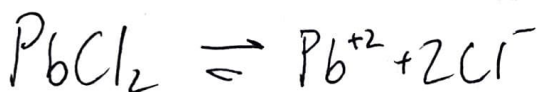
$$= 4S^3 = 4.87 \times 10^{-17}$$

$$S = 2.301 \times 10^{-6} \text{ mol/L}$$

0.500 L	$2.301 \times 10^{-6} \text{ mol/L}$	89.87
	1 L	1 mol
	1.03	$= 2.07 \times 10^{-4} \text{ g}$

6. The molar solubility of PbCl_2 in pure water is $1.43 \times 10^{-2} \text{ M}$ at 25°C.

- a. Write the equilibrium constant expression for the dissolving of $\text{PbCl}_2(\text{s})$.



$$K_{sp} = [\text{Pb}^{+2}][\text{Cl}^-]^2 = 4S^3 = 6$$

- b. How many grams of PbCl_2 can dissolve into 200. mL of pure water?

$1.43 \times 10^{-2} \text{ mol/L}$	288.10 g/mol	
L	mol	

$$= 3.9754 \frac{\text{g}}{\text{L}} \times 0.2 \text{ L}$$

$$= \underline{0.795 \text{ g}}$$

- c. What change could be made to increase amount of PbCl_2 that will dissolve? ↖ not []

add H_2O , heat

Multiple Choice Practice!

S²

7. The solubility product, K_{sp} , of AgCl is 1.8×10^{-10} . Which of the following expressions is equal to the molar solubility of AgCl?

a. $(1.8 \times 10^{-10})^2$ molar

b. $\frac{1.8 \times 10^{-10}}{2}$ molar

c. 1.8×10^{-10} molar

d. $\sqrt{1.8 \times 10^{-10}}$ molar ✓

8. Which of the following is equal to the solubility product, K_{sp} , of Ag_2CO_3 ?

a. $K_{sp} = [\text{Ag}^+][\text{CO}_3^{2-}]$

b. $K_{sp} = [\text{Ag}^+][\text{CO}_3^{2-}]^2$

c. $K_{sp} = [\text{Ag}^+]^2[\text{CO}_3^{2-}]$

d. $K_{sp} = [\text{Ag}^+]^2[\text{CO}_3^{2-}]^2$

9. If the solubility of BaF_2 is equal to x , which of the following expressions is equal to the solubility product, K_{sp} , of BaF_2 ?

a. x^2

b. $2x^2$

c. $2x^3$

d. $4x^3$

10. In a saturated solution of Na_3PO_4 , $[\text{Na}^+] = 0.30 \text{ M}$. What is the molar solubility of Na_3PO_4 ?

a. 0.10 M

b. 0.30 M

c. 0.60 M

d. 0.90 M

11. What is the maximum mass of AgCl can dissolve in 100. mL of pure water at 25°C? The molar solubility of AgCl is 1.3×10^{-5} .

a. $1.9 \times 10^{-4} \text{ g}$

b. $1.9 \times 10^{-5} \text{ g}$

c. $9.1 \times 10^{-9} \text{ g}$

d. $9.1 \times 10^{-6} \text{ g}$

$1.3 \times 10^{-5} \text{ mol/L}$	143.32 g/mol	0.1
L	mol	

The Common Ion Effect

Remember: solubility can change if you change reaction conditions!

- Le Châtelier's principle predicts that a salt will become less soluble in a solution that already contains one of its own ions already dissolved: what's known as a common ion.
- The presence of a common ion acts like increasing the concentration of a product ion in the salt dissolution, causing the system to shift left to establish equilibrium (towards the left side).
(reverse)

Example:

- Circle any of the following compounds that contain a common ion to MgCl_2 :

AgF_2

NaCl

Mg(OH)_2

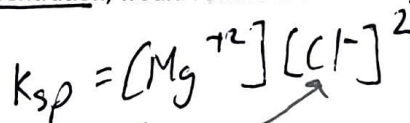
AlCl_3

Al_2S_3

- Which of the compounds above, if present in the solution in equal concentration, would reduce the solubility of MgCl_2 :

a. the most? Why?

AlCl_3 brings 3 Cl^-



in a squared term

b. the least? Why?

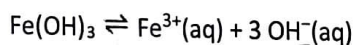
Mg(OH)_2 brings 1 Mg^{2+}

in a single term

The Effect of pH on Solubility

The common ion effect predicts that when a salt contains ions that can act as an acid or a base, the solubility of that salt will be affected by changes in pH.

Example:



- Will iron (III) hydroxide be more, less, or equally soluble in a basic solution (when compared to its solubility in pure water)? Explain.

Basic = OH^- present

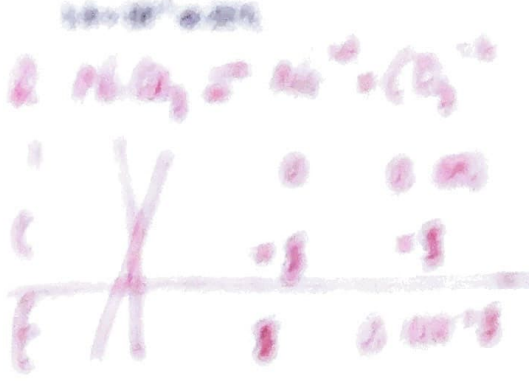
Le Chat says $\rightleftharpoons$ rxn shifts left
b/c of excess product present.

- Will iron (III) hydroxide be more, less, or equally soluble in an acidic solution (when compared to its solubility in pure water)? Explain.

b/c H^+ will react w/ OH^- ,
reducing $[\text{OH}^-]$ in soln & Le Chat
says reduction in product shifts
rxn forward dissolving more
 Fe(OH)_3

Exercise 10: Calculations

1. Calculate the molar solubility of Ag_2S in a solution containing 0.1 M NH_3 and 0.01 M HS^- .



$K_{sp} = (2s)^2(s)$

$1 \times 10^{-50} = (4s^3)(0.01 + s)$

$s = 1 \times 10^{-16} \text{ M}$

$s_{\text{Ag}_2\text{S}} = 1 \times 10^{-16} \text{ M}$

2. Calculate the molar solubility of Ag_2S in a solution containing 0.1 M NH_3 and 0.01 M HS^- and 0.01 M S^{2-} .



$K_{sp} = (2s)^2(s)$

$(4s^3)(0.01 + s) = 1 \times 10^{-50}$

$s = 1 \times 10^{-16} \text{ M}$

Exercise 11: Qualitative Analysis

1. A solution contains Ag^+ , Pb^{2+} , and Cu^{2+} . The solution is treated with HCl and then H_2S in 0.3 M HCl . The precipitates are AgCl , PbCl_2 , and CuS . The solution is then treated with NH_3 and then H_2S in 0.3 M NH_3 . The precipitates are Ag_2S , PbS , and CuS .

2. A solution contains Ag^+ , Pb^{2+} , and Cu^{2+} . The solution is treated with HCl and then H_2S in 0.3 M HCl . The precipitates are AgCl , PbCl_2 , and CuS . The solution is then treated with NH_3 and then H_2S in 0.3 M NH_3 . The precipitates are Ag_2S , PbS , and CuS .



$K_{sp} = (2s)^2(s)$

$(4s^3)(0.01 + s) = 1 \times 10^{-50}$

$s = 1 \times 10^{-16} \text{ M}$

$s_{\text{Ag}_2\text{S}} = 1 \times 10^{-16} \text{ M}$

$s_{\text{PbS}} = 1 \times 10^{-16} \text{ M}$

8. Given a 2.00 L saturated solution of $\text{Cu}(\text{IO}_3)_2$, $K_{sp} = 1.4 \times 10^{-7}$, how many moles of NaIO_3 would need to be dissolved in solution to reduce $[\text{Cu}^{2+}]$ to $6.0 \times 10^{-5} \text{ M}$? (Assume the added NaIO_3 does not appreciably change the total volume of solution.)



$$[\text{IO}_3^-] = 4.8 \times 10^{-2} \frac{\text{mol}}{\text{L}} \times 2.00 \text{ L}$$

$$= 9.7 \times 10^{-2} \text{ moles} \\ 2 \text{ L}$$

$$K_{sp} = [\text{Cu}^{2+}][\text{IO}_3^-]^2 = 1.4 \times 10^{-7}$$

$$(\text{Cu}^{2+})(x+s)^2 = 1.4 \times 10^{-7}$$

$$6.0 \times 10^{-5} (x+s)^2 = 1.4 \times 10^{-7}$$

$$(x+s) = 4.8 \times 10^{-2} \text{ M} \\ = [\text{IO}_3^-]$$

Even more practice!

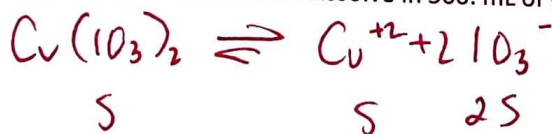
9. Copper(I) bromide has a measured solubility of $2.0 \times 10^{-4} \text{ mol/L}$ at 25°C . Calculate its K_{sp} value.



$$K_{sp} = [\text{Cu}^{+1}][\text{Br}^{-}] = s^2 = 2.0 \times 10^{-4}$$

$$s = 1.4 \times 10^{-2} \text{ M}$$

10. The K_{sp} value for copper(II) iodate, $\text{Cu}(\text{IO}_3)_2$, is 1.4×10^{-7} at 25°C . What is the maximum mass, in grams, of copper(II) iodate that can dissolve in 500. mL of water?



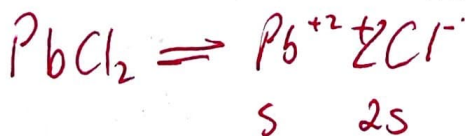
$$K_{sp} = [\text{Cu}^{+2}][\text{IO}_3^-]^2 = 1.4 \times 10^{-7}$$

$$s (2s)^2 = 4s^3 = 1.4 \times 10^{-7}$$

$$s = 3.3 \times 10^{-3} \text{ M}$$

$$3.3 \times 10^{-3} \text{ M} \times 0.5 \text{ L} = 1.64 \times 10^{-3} \text{ mol} \times \frac{413.35 \text{ g}}{1 \text{ mol}} = 0.68 \text{ g} \\ \text{Cu}(\text{IO}_3)_2$$

11. In pure water at 25°C , the molar solubility of PbCl_2 is 1.3×10^{-6} and the K_{sp} is 1.6×10^{-5} . LiCl is added to 5.00 L of a saturated solution of PbCl_2 at 25°C until the $[\text{Pb}^{2+}]$ is reduced to $4.5 \times 10^{-4} \text{ M}$. How many moles of chloride ions are dissolved in solution at the point when $[\text{Pb}^{2+}] = 4.5 \times 10^{-4} \text{ M}$? (Assume the added LiCl has a negligible effect on the total volume of solution.)



$$K_{sp} = [\text{Pb}^{+2}][\text{Cl}^{-}]^2 = 1.3 \times 10^{-6}$$

$$4.5 \times 10^{-4} \cdot (\text{Cl}^{-})^2 = 1.3 \times 10^{-6}$$

$$[\text{Cl}^{-}] = 5.37 \times 10^{-2} \text{ M}$$

Multiple Choice Practice!

Use the following information to answer questions 12–14.

150 mL of saturated SrF_2 solution is present in a 250 mL beaker at room temperature. The molar solubility of SrF_2 at 298 K is $1.0 \times 10^{-3} \text{ M}$.

12. What are the concentrations of Sr^{2+} and F^- in the beaker?

- a. $[\text{Sr}^{2+}] = 1.0 \times 10^{-3} \text{ M}$; $[\text{F}^-] = 1.0 \times 10^{-3} \text{ M}$
- ☒ b. $[\text{Sr}^{2+}] = 1.0 \times 10^{-3} \text{ M}$; $[\text{F}^-] = 2.0 \times 10^{-3} \text{ M}$
- c. $[\text{Sr}^{2+}] = 2.0 \times 10^{-3} \text{ M}$; $[\text{F}^-] = 1.0 \times 10^{-3} \text{ M}$
- d. $[\text{Sr}^{2+}] = 2.0 \times 10^{-3} \text{ M}$; $[\text{F}^-] = 2.0 \times 10^{-3} \text{ M}$

$$K_{sp} = [\text{Sr}^{+2}][\text{F}^-]^2$$
$$1 \times 10^{-3} \cdot 2 \times 10^{-3}$$

13. What would be the effect on $[\text{Sr}^{2+}]$ if some NaF(s) was added to the beaker?

- a. $[\text{Sr}^{2+}]$ would remain unchanged; neither ion in NaF(s) is common to Sr^{2+}
- b. $[\text{Sr}^{2+}]$ would increase; more Sr^{2+} ions would be needed to balance the additional F^- ions to re-establish equilibrium.
- ☒ c. $[\text{Sr}^{2+}]$ would decrease; the additional fluoride ions would cause the system to shift left to re-establish equilibrium.
- d. $[\text{Sr}^{2+}]$ would decrease; the additional Na^+ would cause an excess of positive charge, and the system would shift left to reduce overall positive charge.

14. Calculate the solubility product for SrF_2 at 25°C .

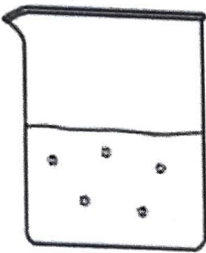
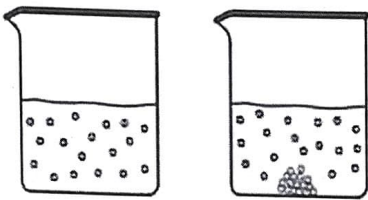
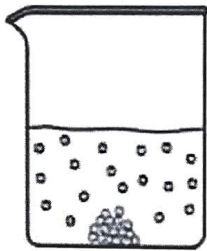
- a. 2×10^{-9}
- ☒ c. 4×10^{-9}
- b. 2×10^{-6}
- d. 4×10^{-6}

$$= 4s^3 = 4 \cdot (1 \times 10^{-3})^3$$

Will a Precipitate Form? A Task for K vs Q!

Precipitation occurs when the concentrations of ions is greater than the solubility of the ionic compound.

Compare the value of Q with given K_{sp} to determine if a precipitate will form!

$K > Q$	$K = Q$	$K < Q$
Unsaturated Solution	Saturated Solution	Saturated Solution with extra
System will shift right to reach equilibrium $\text{AgCl}(s) \rightleftharpoons \text{Ag}^+(aq) + \text{Cl}^-(aq)$	At equilibrium $\text{AgCl}(s) \rightleftharpoons \text{Ag}^+(aq) + \text{Cl}^-(aq)$	System will shift left to reach equilibrium $\text{AgCl}(s) \rightleftharpoons \text{Ag}^+(aq) + \text{Cl}^-(aq)$
More solid will dissolve until $K = Q$.	Solid will both dissolve and precipitate at the same rate.	More solid will precipitate until $K = Q$.
 no precipitate	 maybe precipitate	 yes precipitate

Important Ideas to Note:

- If ANY solid is present, the solution is at equilibrium (a saturated solution)
- Ion concentration, [ions], is **independent** of volume when at equilibrium (for instance, in a saturated solution).
- If ions are present that could form multiple salts, the solid with the smallest molar solubility will form.

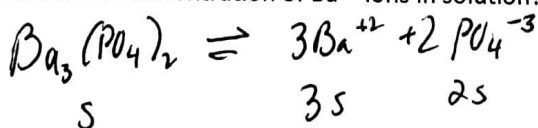
Solubility Equilibrium Translation Guide

- Solubility product constant = K_{sp} (aka the equilibrium constant for solubility)
- Molar solubility = x from RICE table (aka how many moles of a solid will dissolve in 1.0 L, units = M = mol/L)
- Saturated = equilibrium (aka a solution has dissolved as many ions as can fit, any extra will precipitate)

Let's Practice!

- A chemist makes a 2.0 L saturated solution of $\text{Ba}_3(\text{PO}_4)_2$ solution, which has a $K_{sp} = 6.0 \times 10^{-39}$.

- What is the concentration of Ba^{2+} ions in solution?



$$[\text{Ba}^{2+}] = 3s = 3(8.9 \times 10^{-9}) = 2.7 \times 10^{-8} \text{ M}$$

doesn't matter!

$$K_{sp} = [\text{Ba}^{2+}]^3 [\text{PO}_4^{3-}]^2 = 6.0 \times 10^{-39}$$

$$(3s)^3 (2s)^2 = 6.0 \times 10^{-39}$$

$$108s^5 = 6.0 \times 10^{-39}$$

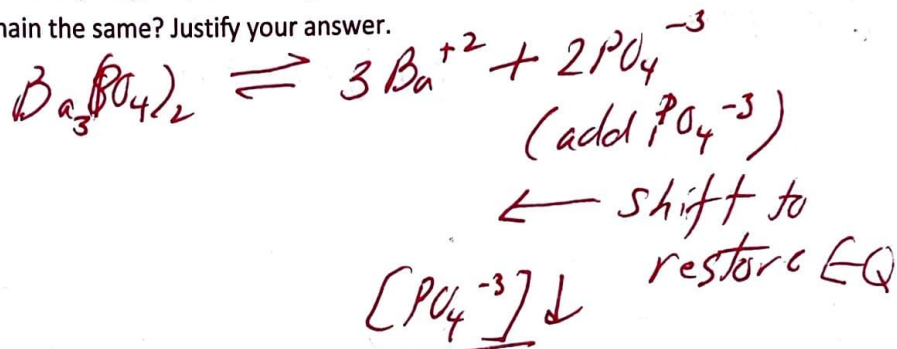
$$s^5 = 5.56 \times 10^{-41}$$

$$s = 8.9 \times 10^{-9}$$

- b. After two days of sitting on the counter, some liquid has evaporated from the solution. Did $[Ba^{2+}]$ increase, decrease, or remain the same? Justify your answer.

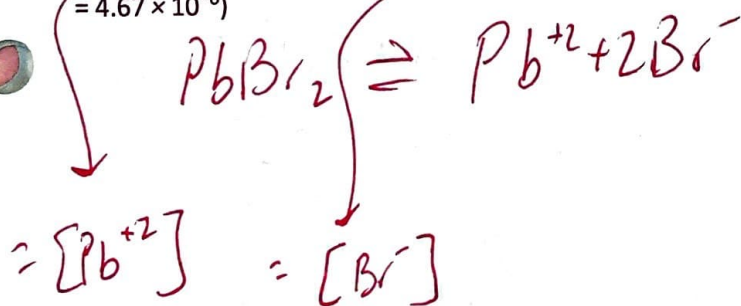
Same! Saturated!

- c. The chemist adds 3.00 g of solid $(NH_4)_3PO_4$ to the original saturated solution of $Ba_3(PO_4)_2$. Did $[Ba^{2+}]$ increase, decrease, or remain the same? Justify your answer.



2. A solution containing lead (II) nitrate is mixed with one containing sodium bromide to form a solution that is 0.0150 M in $Pb(NO_3)_2$ and 0.00350 M NaBr. Does a precipitate form in this newly mixed solution? (K_{sp} of $PbBr_2$

$$= 4.67 \times 10^{-6}$$



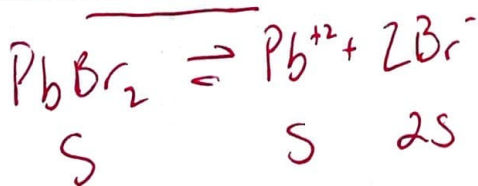
$$K_{sp} = [Pb^{+2}][Br^-]^2 = 4.67 \times 10^{-6}$$

$$Q_{sp} = [Pb^{+2}][Br^-]^2$$

$$= (0.0150)(0.00350)^2 = 1.84 \times 10^{-7}$$

$Q_{sp} < K_{sp}$
unsaturated! No ppt!

3. The K_{sp} value for lead (II) bromide, $PbBr_2$, is 4.6×10^{-6} at $25^\circ C$. What is the maximum mass, in grams, of $PbBr_2$ that can dissolve in 1.50 L of water?



$$K_{sp} = [Pb^{+2}][Br^-]^2 = 4.6 \times 10^{-6}$$

$$S(2S)^2 = 4.6 \times 10^{-6}$$

$$4S^3 = 4.6 \times 10^{-6}$$

$$S = 1.05 \times 10^{-2}$$

$$S = \frac{1.05 \times 10^{-2} M}{1.50 L} = 1.57 \times 10^{-2} mol$$

$$\frac{367g}{1mol} = \boxed{5.8g}$$

5 points

4. A student mixes 15.0 mL of 0.015 M sodium iodide solution, NaI, with 5.00 mL of 0.0025 M $\text{Pb}(\text{NO}_3)_2$. The K_{sp} of PbI_2 is 8.5×10^{-9} . What will the student observe? Justify your answer with calculations.

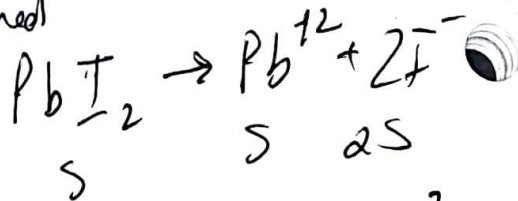
Ugh. Additive volumes when the solns are combined

$$\frac{15 \text{ mL} \cdot 0.015 \text{ M NaI}}{(15+5) \text{ mL}} = 0.01125 \text{ M NaI}$$

$$[I^-] = 0.01125$$

$$\frac{5 \text{ mL} \cdot 0.0025 \text{ M Pb}(\text{NO}_3)_2}{(15+5) \text{ mL}} = 0.002525 \text{ M Pb}(\text{NO}_3)_2$$

$$[\text{Pb}^{2+}] = 0.002525$$

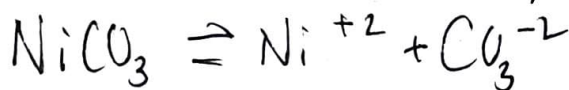


$$Q_{sp} = [\text{Pb}^{2+}][\text{I}^-]^2 = (0.002525)(0.01125)^2 = 3.20 \times 10^{-6} > K_{sp} (8.5 \times 10^{-9})$$

precip will form!

Additive volumes.

5. Sodium carbonate is added to a 0.0024 M solution of the nickel (II) ion. If $[\text{Na}_2\text{CO}_3] = 1.0 \times 10^{-4} \text{ M}$, will a precipitate form? (The K_{sp} of nickel (II) carbonate is 6.6×10^{-9} .)

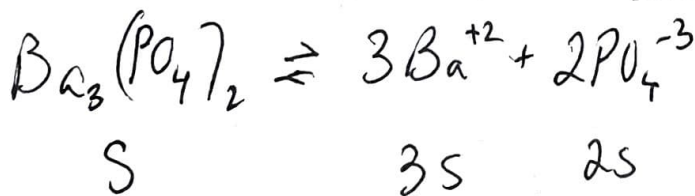


$$Q_{sp} = [\text{Ni}^{2+}][\text{CO}_3^{2-}] = (0.0024)(1.0 \times 10^{-4}) = 2.4 \times 10^{-7}$$

$$Q_{sp} > K_{sp} = 6.6 \times 10^{-9}$$

PPT forms!

6. Calculate the molar solubility of $\text{Ba}_3(\text{PO}_4)_2$, which has a $K_{sp} = 6.0 \times 10^{-39}$.



$$K_{sp} = [\text{Ba}^{2+}]^3 [\text{PO}_4^{3-}]^2 = (3s)^3 (2s)^2 = 108s^5 = 6.0 \times 10^{-39}$$

$$s = 8.9 \times 10^{-9} \text{ M}$$

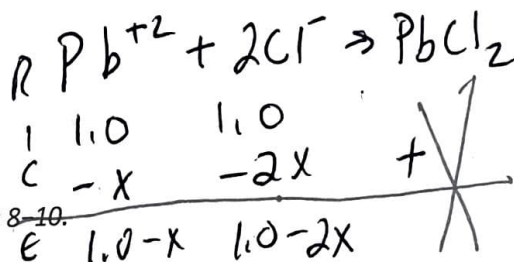
Multiple Choice Practice!

7. 150 mL of saturated SrF_2 solution is present in a 250 mL beaker at room temperature. If some of the solution evaporates overnight, which of the following will occur?

- a. The mass of the solid and the concentration of the ions will remain the same.
- b. The mass of the solid and the concentration of the ions will increase.
- ☒ c. The mass of the solid will decrease, and the concentration of the ions will remain the same.
- d. The mass of the solid will increase, and the concentration of the ions will remain the same.

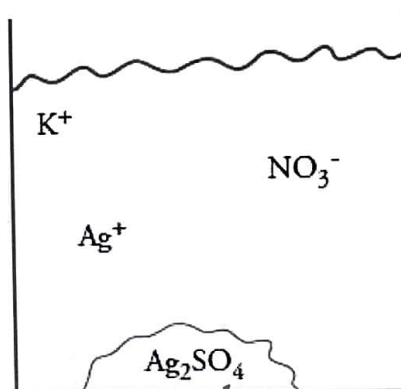
8. A student added 1 liter of a 1.0 M KCl solution to 1 liter of a 1.0 M $\text{Pb}(\text{NO}_3)_2$ solution. A lead chloride precipitate formed, and nearly all of the lead ions disappeared from solution. Which of the following lists the ions remaining in the solution in order of decreasing concentration?

- a. $[\text{NO}_3^-] > [\text{K}^+] > [\text{Pb}^{2+}]$
- b. $[\text{NO}_3^-] > [\text{Pb}^{2+}] > [\text{K}^+]$
- c. $[\text{K}^+] > [\text{Pb}^{2+}] > [\text{NO}_3^-]$
- d. $[\text{K}^+] > [\text{NO}_3^-] > [\text{Pb}^{2+}]$



Use the following information to answer questions 8-10.

Silver sulfate, Ag_2SO_4 , has a solubility product constant of 1.0×10^{-5} . The diagram to the right shows the products of a precipitation reaction in which some silver sulfate was formed.



9. What is the identity of the excess reactant?

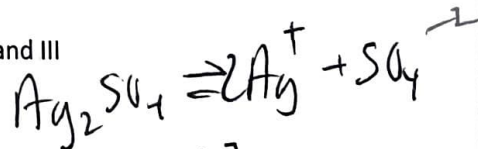
- a. AgNO_3
- b. Ag_2SO_4
- c. NaNO_3
- d. Na_2SO_4

10. If the beaker above was left uncovered for several hours:

- I. Some of the Ag_2SO_4 would dissolve.
 - II. Additional Ag_2SO_4 would precipitate.
 - III. $[\text{Ag}^+]$ would remain constant.
- a. I only b. II only ☒ c. II and III d. I and III

11. Which ion concentration below would have led the precipitate to form?

- a. $[\text{Ag}^+] = 0.01 \text{ M}, [\text{SO}_4^{2-}] = 0.01 \text{ M}$
- ☒ b. $[\text{Ag}^+] = 0.10 \text{ M}, [\text{SO}_4^{2-}] = 0.01 \text{ M}$
- c. $[\text{Ag}^+] = 0.01 \text{ M}, [\text{SO}_4^{2-}] = 0.10 \text{ M}$
- d. It is impossible to determine without knowing the total volume of the solution.



Handwritten Ksp expression for question 11:

$$K_{sp} = [\text{Ag}^+]^2 [\text{SO}_4^{2-}]$$

Handwritten calculation for question 11:

$$K_{sp} = 1 \times 10^{-5}$$

Solubility Summary Sheet

Solubility Language	Normal Equilibrium Language
solubility product constant (K_{sp})	equilibrium constant (K)
molar solubility	x (from RICE table)
saturated solution	system at equilibrium

When you need to solve for molar solubility

Given?	Asked to find?	Strategy
K_{sp}	molar solubility	- Write K_{sp} expression using ions produced when solid dissolves. - Substitute x values from mini-RICE table - Solve for x
K_{sp} OR - Concentration of all ions in a saturated solution	# of grams that can dissolve	- Write K_{sp} expression using ions produced when solid dissolves. - Substitute x values from mini-RICE table - Solve for x (in M = mol / L) - Use the volume of solution to calculate moles that can dissolve - Use molar mass to convert to grams

When you don't need to solve for molar solubility

Given?	Asked to find?	Use:
Concentration of all ions in a saturated solution	K_{sp}	- Write K_{sp} expression using ions produced when solid dissolves. - Substitute given concentrations and solve for K_{sp}.
other ion concentrations at equilibrium <u>and</u> K_{sp}	Concentration of ONE ion in a saturated solution	- Write K_{sp} expression using ions produced when solid dissolves. - Plug in known values, solve for unknown concentration.
ion concentrations when solutions are added or mixed <u>and</u> K_{sp}	If a precipitate will form	- Write Q_{sp} expression using ions produced when solid dissolves. - Substitute given concentrations and solve for Q_{sp}. - Compare K and Q: $K > Q$ = no precipitate $K < Q$ = yes precipitate